

A Variable Granularity Search-Based Multiobjective Feature Selection Algorithm for High-Dimensional Data Classification

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Abstract—Evolutionary algorithms (EAs) have shown their competitiveness in solving the problem of feature selection (FS). However, in most of the existing EA-based FS methods, one bit in the individual only represents one feature, which means with the number of features increasing, the search space of these methods increases exponentially and makes them not suitable for the data classification with high dimensions. To tackle the issue, in this article, a variable granularity search-based multiobjective EA, termed as VGS-MOEA, is proposed for high-dimensional FS, where one bit in the individual representation denotes a group of features and results in the search space reducing greatly. To be specific, at the beginning, the search granularity of VGS-MOEA is coarse (a bit denotes a great number of features), which helps the proposed algorithm detect the potentially good feature subsets quickly. As the evolution continues, the search granularity is refined gradually, where a bit denotes a smaller number of features until it only represents one feature. With this decomposition of granularity, a more refined search is performed and leads to the VGS-MOEA obtaining feature subsets with higher quality. Experimental results on 12 high-dimensional data sets with different characteristics have shown that in comparison with the state of the arts, the proposed VGS-MOEA has demonstrated its superiority in terms of the classification accuracy, the number of selected features, and the running time.

Index Terms—Evolutionary algorithm (EA), feature selection (FS), high-dimensional data classification, multiobjective optimization.

I. INTRODUCTION

DATA classification is an important topic in machine learning area, where the labeled data is utilized to train a

classifier with high accuracy [1]. Usually, the training data is described by a group of features, which may contain some irrelevant or redundant information. To this end, feature selection (FS) as a data preprocessing technology aims to remove these irrelevant or redundant features and obtain a feature subset with similar or even better classification performance [2]. Due to the merits above, in the last two decades, a great deal of FS algorithms with different techniques have been proposed [2]–[5]. Among them, the evolutionary computing (EC)-based methods have gained much focus, since they have the ability of global search and do not need domain knowledge or assumption on the data distribution [2]. The existing EC-based FS algorithms are mainly divided into three categories: 1) the filter-based methods [6], [7]; 2) the wrapper-based methods [8], [9]; and 3) the embedding-based methods [10], [11]. Among them, the wrapper-based methods have attracted much focus, since they achieve high-quality feature subsets by using a classifier (e.g., KNN and SVM) as the evaluation tool. Therefore, in this article, we also concentrate on the EC-based wrapper FS algorithms.

According to the EC paradigms they used, the existing EC-based wrapper FS algorithms include genetic algorithm (GA)-based approaches [12], [13], particle swarm optimization (PSO)-based approaches [14]–[16], genetic programming (GP)-based approaches [17]–[19], and so on. These evolutionary algorithms (EAs) have shown their competitiveness in achieving good feature subsets. However, most of them are not suitable for the high-dimensional situations, which often occur in the real FS applications, such as gene analysis [20] and image recognition [21]. To tackle the issue, recently, many efforts have been made to develop high-dimensional FS algorithms, and most of them adopt the PSO as the framework, since the PSO has the advantages of easy implementation and fast convergence [8], [22], [23].

For example, Banka and Dara [24] proposed a Hamming distance-based binary PSO (HDBPSO) algorithm for high-dimensional FS, where the Hamming distance was utilized as a proximity measure to update the velocity of particle(s) in binary PSO. Experimental results on high-dimensional data sets have shown that HDBPSO could obtain the feature subset with high quality. Tran *et al.* [8] suggested a variable-length PSO (VLPSO) for FS with high dimensions. In VLPSO, the particles had different and shorter lengths, which led the search space of VLPSO to be greatly reduced and made it suitable for the high-dimensional data classification. Xue *et al.* [25]

Manuscript received 1 June 2021; revised 4 September 2021, 7 December 2021, and 30 January 2022; accepted 8 March 2022. Date of publication 18 March 2022; date of current version 31 March 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62076001, Grant 61976001, Grant 61876184, and Grant U1804262; in part by the Key Projects of University Excellent Talents Support Plan of Anhui Provincial Department of Education under Grant gxyqZD2021089; in part by the Natural Science Foundation of Anhui Province under Grant 2008085QF309; in part by the Humanities and Social Sciences Project of Chinese Ministry of Education under Grant 18YJC870004; and in part by the University Synergy Innovation Program of Anhui Province under Grant GXXT-2020-050. (Corresponding author: Lei Zhang.)

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This article has supplementary material provided by the authors and color versions of one or more figures available at <https://doi.org/10.1109/TEVC.2022.3160458>.

Digital Object Identifier 10.1109/TEVC.2022.3160458

designed two self-adaptive PSO algorithms, termed as SaPSO and SPS-PSO [26] for large-scale FS. Different from other PSO-based algorithms that use only one candidate solution generation strategy (CSGS), the designed algorithms maintained multiple CSGSs, from which different self-adaptive mechanisms were developed to choose the suitable CSGSs to generate new particles. Song *et al.* [23] proposed a variable-size cooperative coevolutionary PSO algorithm (VS-CCPSO), where the idea of “divide and conquer” was utilized to tackle the challenge of high-dimensional FS. Besides these variants of PSO, we note that there also exist a few high-dimensional FS algorithms with other evolutionary frameworks [27]–[29].

Despite these EA algorithms have shown their efficiency and effectiveness in tackling the FS problem with high dimensions. However, by taking a close look at these algorithms, we can find that most of them still adopt the traditional encoding scheme (e.g., 0-1 encoding or real encoding), which means each bit in the individual represents only one feature, and their search granularity is one (denotes one feature). During the evolution, the search space of the individuals in these EAs is $O(2^n)$, which is very huge if the number of total features (n) is large. To handle the “curse of dimensionality,” these high-dimensional FS algorithms, especially the PSO variants develop various problem-guided operators or strategies to enhance the search efficiency. Different from most of the existing ones, which mainly focus on the PSO framework, this article aims to provide a general EA framework for high-dimensional FS problem. To be specific, in this article, a variable granularity search-based MOEA, named VGS-MOEA, is suggested. In VGS-MOEA, each bit in the individual denotes a group of features, which makes the search granularity of VGS-MOEA is much bigger than that of the existing EAs and results in the search space of our individuals being greatly reduced. In addition, as the evolution proceeds, the search granularity becomes smaller, which helps the search operation of the VGS-MOEA be more refined, and finally, find the feature subsets with high quality. In summary, the main contributions of this work can be described as follows.

- 1) A variable granularity search (VGS) scheme is suggested for EA-based high-dimensional FS. In VGS, at the beginning of evolution, one bit in the individual encoding represents a great number of related features, thus, the search granularity is large (coarse granularity search), with which the potentially good feature subsets can be quickly found. As the evolution continues, the coarse granularity is decomposed gradually, which means one bit denotes a smaller number of features until in the end one bit denotes one feature. Through the granularity changing, the refined search is carried on, and finally obtains the feature subsets with higher quality.
- 2) Based on the suggested VGS, a multiobjective FS algorithm named VGS-MOEA is proposed for high-dimensional data classification, where the VGS is applied to tackle the curse of dimensionality. In VGS-MOEA, two variable granularity-based genetic operators (e.g., crossover and mutation) are designed, which overcome the challenge caused by different sizes of granularity. In addition, a granularity-based archive strategy

is also developed, with which the performance of VGS-MOEA is further improved.

- 3) The efficiency and the effectiveness of the proposed VGS-MOEA are validated on 12 different classification data sets, whose feature numbers range from 1024 to 21 655. Experimental results show that in comparison with the state of the arts, the VGS-MOEA has better performance in terms of the classification accuracy, the number of selected features, and the running time.

The remainder of this article is organized as follows. The related work on the EC-based FS algorithms is reviewed in Section II. Then, we present the details of the proposed VGS-MOEA in Section III. The effectiveness and the efficiency of VGS-MOEA are verified in Section IV, by comparing it with several state-of-the-art EC-based high-dimensional FS algorithms. Finally, we provide our conclusions and discuss the future work in Section V.

II. RELATED WORK

In this section, we first introduce the related work on the EC-based FS algorithms. Then, we present the recent progress on the EC-based high-dimensional FS algorithms, which is closely related to the topic of this article.

A. EC-Based Feature Selection Algorithms

In the last two decades, as an optimization technique, evolutionary computation has attracted much attention due to its advantages of global search ability and no assumptions on the data space. Correspondingly, a great deal of EAs with different EC frameworks have been proposed for the task of FS, whose aim is to select a small number of features with similar (or higher) accuracy from the original features.

For example, Tan *et al.* [12] suggested a GA-based approach for FS, where GA was combined with multiple FS criteria to achieve feature subset with smaller size and higher classification accuracy. Similarly, Too and Abdullah [13] developed a new and fast GA for FS, which utilized the competition strategy that combined the new selection and crossover schemes to obtain competing solutions quickly. Zhang *et al.* [17], Tran *et al.* [18], and Nag and Pal [19] designed different GP-based FS algorithms. Experimental results on various FS applications (e.g., job-shop scheduling and disease diagnosis) have demonstrated the effectiveness of their algorithms. PSO, as a swarm intelligence optimization technique, has also been successfully utilized to tackle the issue of FS, and a series of PSO-based studies have been conducted [8], [14], [30], where different initialization strategies, fitness functions, and search mechanisms were designed to obtain high-quality feature subsets. Other FS algorithms with various EC paradigms can be found in [31] and [32].

These EAs have shown their competitiveness in obtaining the feature subsets with high quality, however, most of them use single-objective optimization, which means they need to integrate two FS objectives (the classification accuracy and the number of selected features) into one objective. To this end, a tradeoff parameter is often introduced to combine two objectives. A fact is that how to set a suitable value for this

parameter is a difficult problem, especially when there is no prior knowledge be obtained. To tackle the problem, a series of multiobjective EAs (MOEAs) have been proposed for FS.

For instance, Nguyen *et al.* [9] suggested a multiple reference points-based decomposition method for multiobjective FS, where a static mechanism (MOEA/D-STAT) and a dynamic mechanism (MOEA/D-DYN) were designed to overcome the challenges of the highly discontinuous Pareto front and imbalance preferences in the FS task. To be specific, the static mechanism alleviated the dependence of the decomposition on the Pareto front shape and the effect of the discontinuity. While the dynamic one was able to detect regions in which the objectives are mostly conflicting, and allocated more computational resources to the detected regions. Empirical studies on 12 classification data sets have shown that compared with the existing EA-based FS algorithms, their suggested method had higher performance in terms of accuracy, the number of selected features, and the HV values. In addition, the comparison results have also revealed that the dynamic mechanism was better than the static one.

Xue *et al.* [16] presented the first study on multiobjective PSO for FS, where two PSO-based multiobjective FS algorithms were proposed in their study. The first algorithm introduced the idea of nondominated sorting into PSO to address the FS problem. The second algorithm applied the ideas of crowding, mutation, and dominance to PSO to search for the Pareto front solutions. The comparison results have justified the effectiveness of their PSO-based MOEAs. Following this work, more multiobjective PSO-based FS algorithms have been developed [33], [34].

Hancer *et al.* [35] suggested a Pareto front FS algorithm, named MOABC, where artificial bee colony (ABC) was adopted as the optimization framework. To be specific, a new multiobjective ABC algorithm that integrated with nondominated sorting and genetic operators was suggested. Based on it, two different implementations were developed: 1) Bin-MOABC (binary version) and 2) Num-MOABC (continuous version). The empirical results have verified the efficiency and effectiveness of MOABC. In addition, the results also indicated that Bin-MOABC achieved better feature subsets than Num-MOABC.

Zhang *et al.* [36] proposed a binary differential evolution with self-learning strategy, named MOFS-BDE, for FS. In MOFS-BDE, three new operators were designed to improve the performance of their algorithm. Specifically, the novel binary mutation operator based on probability difference can guide individuals to rapidly locate potentially optimal areas. The one-bit purifying search operator can improve the self-learning capability of the elite individuals located in the optimal areas. The efficient nondominated sorting operator with crowding distance can reduce the computational complexity of the selection operator in the differential evolution.

Recently, Xu *et al.* [37] designed a duplication analysis-based EA (DAEA) for biobjective FS. In DAEA, three important improvements were proposed, which includes: 1) a reproduction process that can improve the quality of offspring; 2) a duplication analysis method that removes the redundant solutions; and 3) a diversity-based selection strategy that further

selects the reserved solutions. Experimental results on different FS tasks have demonstrated the effectiveness of DAEA. More MOEAs for FS can be found in [2], [4], [35], and [37]–[39].

B. EC-Based Feature Selection Algorithms for High-Dimensional Data Classification

The EC-based algorithms mentioned above have shown their competitiveness in solving the FS problem. However, in some real FS applications, the number of features in the data is very large, which brings a great challenge to the existing EC-based FS algorithms. To tackle the challenge, recently, many efforts have been made to design EC-based algorithms for high-dimensional FS, and a series of representative works are the algorithms based on the framework of PSO.

For example, Banka and Dara [24] proposed an HDBPSO algorithm for high-dimensional FS. In HDBPSO, the initially high-dimensional data set was processed by using a quartile-based fast heuristic technique to reduce the crude domain features. Then, the Hamming distance was introduced as a proximity measure to update the velocity of particle(s) in the binary PSO framework and select the important feature subsets. The experimental results on the high-dimensional gene expression data sets have demonstrated the effectiveness of HDBPSO.

Tran *et al.* [8] suggested a VLPSO representation for FS on high-dimensional classification. Different from the existing PSO-based algorithms for FS, which use the fix-length representation, the suggested VLPSO enables particles to have different and shorter lengths, with which the search space of VLPSO is greatly reduced. To be specific, in their suggested method, the comprehensive learning PSO (CLPSO) with an adaptive learning probability was adopted as the basic optimizer. Then, the population division, feature ranking, and length changing strategies were enabled by the variable length (VL) representation were developed. In addition, a hybrid evaluation method that combines the KNN and a distance measure was also designed, which combined the strengths of the wrapper and filter methods. Results on ten high-dimensional data sets have shown that VLPSO achieved smaller feature subsets with significantly higher classification performance in much shorter time than the fixed-length PSO algorithms as well as other non-PSO FS algorithms.

Song *et al.* [23] designed a VS-CCPSO algorithm, where the idea of divide and conquer was employed to solve the problem of high-dimensional FS. To be specific, a space division strategy based on the feature importance was first presented, which can classify relevant features into the same subspace with a low computational cost. Then, an adaptive adjustment mechanism of subswarm size was developed to maintain an appropriate size for each subswarm, with the purpose of saving computational cost on evaluating particles. Finally, a particle deletion strategy and a particle generation strategy were also designed, which can further enhance the performance of VS-CCPSO. Experimental results on various data sets have shown its competitiveness for tackling FS problems with high dimensions. Following this work, in 2021, Song *et al.* [40] further

proposed a fast hybrid FS based on correlation-guided clustering and PSO (HFS-C-P) for high-dimensional classification. Specifically, three phases of FS methods were effectively integrated into the proposed algorithm based on their respective advantages. Moreover, a symmetric uncertainty-based feature deletion method, a fast correlation-guided feature clustering strategy, and an improved integer PSO were developed to improve the performance of the three phases, respectively.

Lately, inspired by multifactorial optimization (MFO), an evolutionary multitasking-based FS method named PSO-EMT was developed for high-dimensional classification [41]. In PSO-EMT, two related tasks about the target concept were established by evaluating the importance of features. A new crossover operator was applied to share information between these two related tasks. In addition, two mechanisms (variable-range mechanism and subset updating mechanism) were also developed to reduce the search space and maintain the diversity of the population, respectively. The results on the examined high-dimensional classification problems have shown that the proposed PSO-EMT can achieve higher classification accuracy with a smaller feature subset in a reasonable time than the state of the arts.

Despite these PSO-based algorithms have shown their effectiveness in solving the FS with high dimensions, all of them adopt single-objective optimization. In the literature of multiobjective optimization, to the best of our knowledge, there are few works, and the most related one is the SM-MOEA proposed by Cheng *et al.* [27]. In SM-MOEA, a steering matrix was first designed according to the importance of the feature in the data set and the individual in the population. Based on the designed matrix, two important operators referred as dimensionality reduction and individual repairing operators were developed, with which not only the search efficiency of SM-MOEA was improved greatly but also it obtained the feature subsets with higher quality. In addition, we note that since the problem of high-dimensional FS is a kind of sparse large-scale multiobjective problems (Sparse LSMOPs), thus, some recently suggested MOEAs (e.g., SparseEA [28] and MOEA/PSL [29]) for Sparse LSMOPs can be applied to solve the high-dimensional FS problem efficiently.

From the reviews above, we can find that most of the existing EAs for high-dimensional FS solve the curse of dimensionality by developing various problem-guided operators or strategies to reduce the huge search space, however, their search granularity is stable and set as one (one bit represents one feature). Different from these EAs, in the following, we will introduce a variable granularity search-based MOEA (VGS-MOEA), where different sizes of search granularity are carried during the evolution. It ensures the VGS-MOEA achieves higher classification accuracy with a smaller feature subset in a much less running time. In the following, we will discuss the proposed VGS-MOEA in detail.

III. PROPOSED METHOD VGS-MOEA

In this section, we present the details of the proposed VGS-MOEA for high-dimensional FS. First, we present the suggested VGS scheme (including the basic idea, granularity initialization, and the granularity refinement), which is the key

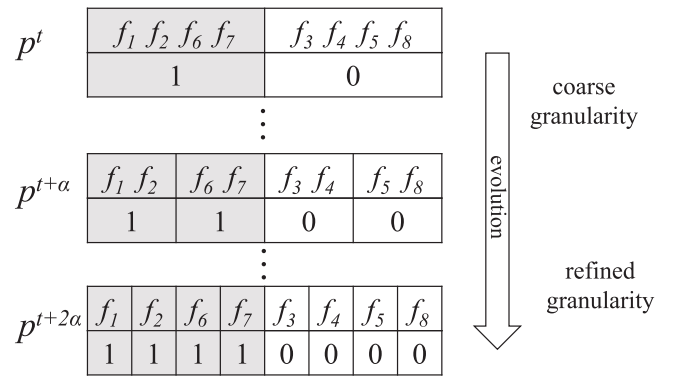


Fig. 1. Illustrative example of the encoding scheme of VGS with eight features, where p^t is the individual at the t -th generation. α is the number of generations between the two conversions, which changes the individuals from coarse granularity to refined granularity.

component of the proposed method. Then, for the problem that traditional genetic operations (e.g., crossover and mutation) are not suitable for the individuals with different granularity in VGS-MOEA, two granularity-based genetic operations are also given. Third, to further improve the performance of the proposed method, an archive strategy based on the variable granularity scheme is suggested. Finally, we present the general framework of the proposed VGS-MOEA.

A. Variable Granularity Search Scheme

1) *Basic Idea:* As mentioned before, most of the existing EA-based FS algorithms adopt one bit to represent a feature. Thus, if the number of total features is large, the search space of their individuals is very huge, which brings a great challenge for the existing algorithms. To this end, a variable granularity search scheme named VGS is suggested. The basic idea of VGS can be described as follows. For the high-dimensional FS problem, there are many irrelevant or redundant features, therefore, at the beginning of search, the search granularity is set coarse (one bit represents a group of features), with which the search space is reduced to a very low scale, and the potentially good feature subsets can be quickly found. Then, as the evolution continues, the search granularity is decomposed to a smaller size until the final granularity equals to 1 in the end. At that time, each bit only represents one feature, which ensures that each feature is considered in the evolution. By using the procedure that starts from a large feature group then ends with a single feature, the proposed VGS guarantees both the search efficiency and the quality of obtained feature subsets.

Specifically, the suggested VGS scheme adopts the binary encoding as the individual representation. Different from the existing one, in the VGS encoding scheme, one bit i represents a group of features. If $i = 1$, it means this group of features is all selected, and if $i = 0$, it indicates all the features are not selected. The so-called variable granularity denotes the number of features represented by one bit is changing from large to small during the evolution, with which the search operation is performed from coarse-granularity to refined-granularity gradually. Fig. 1 gives an illustrative example to show the encoding scheme of VGS with eight features. Assume that there is an

individual p^t , and at the beginning, the granularity of p^t is coarse, and each bit in p^t represents four features. Thus, for the total eight features, individual p^t is denoted by two bits. Among them, the first bit represents $\{f_1, f_2, f_6, f_7\}$ and the second bit represents $\{f_3, f_4, f_5, f_8\}$. The value of first bit is 1, which means features f_1, f_2, f_6, f_7 are all selected in individual p^t . While the value of second bit is 0, it indicates features f_3, f_4, f_5, f_8 are not included in p^t . Then, with the evolution continues (e.g., after α generations), the search granularity becomes small, each bit in current individual $p^{t+\alpha}$ denotes two features, and search operation is more refined than that of p^t . After another iterations, the search granularity is decomposed smaller, and at that time one bit in individual $p^{t+2\alpha}$ represents one feature, which ensures every feature in original data is considered, and guarantees the finally obtained feature subset with high quality.

From the example in Fig. 1, we can find that by using the suggested VGS scheme, the search space is greatly reduced, and the search efficiency is improved significantly. In addition, as the evolution goes on, the search granularity is becoming smaller gradually until one bit represents only one feature. These refined-granularity searches ensure that each feature in the data is considered and finally achieves good solutions. However, this example also shows that to implement the suggested VGS scheme, there are two issues to be tackled.

- 1) At the beginning of evolution (coarse-granularity search), which features can be grouped together and be represented by one bit?
- 2) As the evolution continues, how to decompose the search granularity and perform the refined-granularity search gradually.

In the following two sections, we will tackle these two issues in detail.

2) *Coarse-Granularity Initialization*: To tackle the first issue, a feature clustering method is designed to initialize the coarse granularity, where the features belong to the same cluster will be represented by one bit. To be specific, in this work, a feature f is characterized by a vector $\langle SU_f, PCC_f \rangle$. The SU_f denotes the symmetric uncertainty, which is used to measure the nonlinear relationship between feature f and the label [8]. The PCC_f is the Pearson correlation coefficient, which is adopted to capture the linear relationship between feature f and the label [42]. We note that we use the vector $\langle SU_f, PCC_f \rangle$ instead of other simple metrics (e.g., mutual information) lies in the fact that the solutions obtained by the former are better than those of the latter.

Based on the vector representation $\langle SU_f, PCC_f \rangle$, we utilize a clustering algorithm (e.g., K -Means [43]) to cluster all the features into k groups, then, the features that are clustered in the same group will be represented by one bit. In this way, the search space is greatly reduced from $O(2^n)$ to $O(2^k)$ ($k \ll n$, where n is the number of all features in the data). The specific procedure of the coarse-granularity initialization is described in Algorithm 1.

Fig. 2 further presents this procedure with eight features as an example. First, we calculate the SU and PCC values of each feature, which are depicted in the left table of Fig. 2. For example, f_1 is represented as a vector $\langle 0.82, 0.30 \rangle$, f_2 is represented

Algorithm 1 GranularityInitialization(*data*, *label*, *k*)

Input: *data*: the data set; *label*: the labels; *k*: the group number;

Output: *G*: the initialized coarse granularity;

- 1: $SU_f \leftarrow$ Get the normalized symmetric uncertainty for all features in *data*;
 - 2: $PCC_f \leftarrow$ Get the normalized Pearson correlation coefficient for all features in *data*;
 - 3: $g_1, g_2, \dots, g_k \leftarrow KMeans(\langle SU_f, PCC_f \rangle, k)$;
 - 4: $G = \{g_1, g_2, \dots, g_k\}$;
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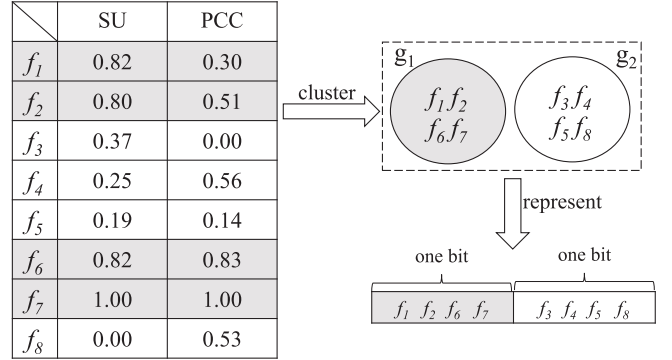


Fig. 2. Example of coarse-granularity initialization.

as $\langle 0.80, 0.51 \rangle$, and so on. Then, the clustering method (e.g., K -Means) is used to group eight features (vectors) into two groups (assuming $k = 2$), where f_1, f_2, f_6, f_7 are clustered in one group g_1 , and f_3, f_4, f_5, f_8 are grouped in another group g_2 (see the upper right part of Fig. 2). Therefore, the results of the coarse-granularity initialization is that the features f_1, f_2, f_6, f_7 and f_3, f_4, f_5, f_8 are represented by one bit, respectively, which are shown in the lower right part of Fig. 2.

3) *Granularity Refinement*: To solve the challenge that how to perform the refined-granularity search during the evolution, a simple yet effective granularity refinement is suggested and applied on all the individuals in the population, where one coarse granularity is decomposed into two refined granularity with the same size, until one bit represents only one feature in the end. Suppose that the granularity of the i th bit in individual p is $G_p(i)$, which means that the i th bit in p denotes $G_p(i)$ features. After evolving for certain generations, its search granularity is refined, and the two refined granularities are $\lceil G_p(i)/2 \rceil$ and $\lfloor G_p(i)/2 \rfloor$. Specifically, $\lceil G_p(i)/2 \rceil$ features are randomly selected from the original $G_p(i)$ features to construct one new bit, and the remained ones are represented by another new bit. Note that the values of two bits in the refined granularity individual inherit from the corresponding bit in the original coarse-granularity individual. If the encoding value of one bit in the original individual is 1, then the values in two refined individuals are also set as 1, vice versa.

We take the individual p^t shown in the top part of Fig. 1 as an example. The first bit represents features f_1, f_2, f_6, f_7 and its granularity is four. After α generations, the coarse granularity is decomposed as two, which indicates f_1, f_2 and f_6, f_7 are represented by one bit in the individual $p^{t+\alpha}$, respectively. Since the original value of bit in individual p^t is 1, thus, the

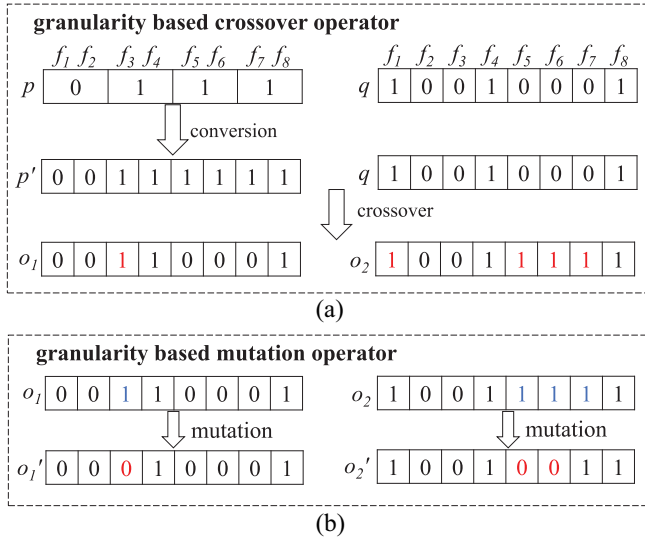


Fig. 3. Example of the suggested granularity-based operators. (a) Granularity-based crossover operator. (b) Granularity-based mutation operator.

decomposed two bits $\{f_1, f_2\}$, $\{f_6, f_7\}$ in $p^{t+\alpha}$ are set as 1. Similarly, the bit $\{f_3, f_4, f_5, f_8\}$ in p' is decomposed as two bits $\{f_3, f_4\}$ and $\{f_5, f_8\}$, whose values are both 0 in individual $p^{t+\alpha}$. By utilizing the decomposition operations above, the coarse-grained individual $p^t = (1, 0)$ is refined into a fine-grained individual $p^{t+\alpha} = (1, 1, 0, 0)$.

B. Granularity-Based Genetic Operators

With the suggested VGS scheme, we can solve the curse of dimensionality in the high-dimensional FS effectively. However, it should be admitted that VGS also brings some new challenges. For example, the traditional genetic operators (e.g., crossover and mutation) are not suitable, since the individuals in the mating pool may have different sizes of granularity. In addition, in the high-dimensional FS problem, the number of final selected features is very small compared with the original numbers. Thus, how to generate sparse offspring under variable granularity is another challenge to be solved. To tackle the challenges above, two granularity-based genetic operators (i.e., granularity-based crossover and granularity-based mutation) are developed, where the former operation is used for exploration, while the latter one is for exploitation.

1) Granularity-Based Crossover Operator: For two individuals with different sizes of granularity, the coarse-granularity individual is first converted into the individual, which has the same granularity as the refined-granularity one. Specifically, suppose individual p is a coarse-granularity one and q is a refined one. In the process of conversion, we adopt the granularity refinement strategy that suggested in Section III-A3 to refine the coarse-granularity individual p repeatedly, until it has the same granularity as q . By doing so, we ensure that the individual after refined (denotes as p') has the same feature group as q . The value of each bit in p' inherits from the original value of p . We take two individuals $p = (0, 1, 1, 1)$ and $q = (1, 0, 0, 1, 0, 0, 0, 1)$ shown in the top part of Fig. 3(a) as an example. Obviously, p is the coarse-granularity individual and each bit's granularity is two, q is the refined-granularity

Algorithm 2 GranularityBasedCrossover(p, q, p_c)

Input: p : the coarse-granularity parent; q : the refined-granularity parent; p_c : crossover probability;

Output: o_1, o_2 : two offsprings;

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1:  $p' \leftarrow$  Convert  $p$  from coarse-individual to refined-individual;
2: if  $\text{rand} < p_c$  then
3:    $o_1 = p' \wedge q$ ; // Case1, Case2
4:    $o_2 = p' \wedge q$ ;
5:    $B_c \leftarrow$  Get bits just selected by  $p$  or  $q$ ; // Case 3
6:    $m \leftarrow$  Get the size of  $B_c$ ;
7:    $m' \leftarrow$  Get a random number between 1 and  $m$ ;
8:   for  $i = 1$  to  $m$  do
9:      $j \leftarrow$  Get the encoding index of  $i$ -th feature in  $B_c$ 
10:    if  $\text{rand} < \frac{m'}{m}$  then
11:       $o_1(j) = 1$ ;
12:    else
13:       $o_2(j) = 1$ ;
14:    end if
15:  end for
16: end if

```

individual and each bit's granularity is one. Therefore, by using the suggested granularity refinement strategy, the granularity of individual p is decomposed into $\lceil 2/2 \rceil$ and $\lfloor 2/2 \rfloor$, which equals to 1. Thus, we obtain the converted individual p' . At this time, the p' already has the same granularity as q . It means that p' does not need to be refined furthermore, and the crossover operation can be performed. In addition, since the first bit value in p is 0, after the conversion, the corresponding values of two bits $\{f_1\}$, $\{f_2\}$ in p' are 0, 0. Similarly, the second bit value in p is 1, the corresponding values of two bits $\{f_3\}$, $\{f_4\}$ in p' are 1, 1, and so on.

When the granularity conversion is finished, the designed crossover operator can be performed on them. The basic idea of the designed crossover is from the following two intuitions: 1) the more important the feature in the parent individuals, the higher probability it will be selected in the offspring individuals and 2) the solutions for the high-dimensional FS should be sparse. Based on two intuitions above, we design the crossover operator by considering the following three cases.

Case 1: The bit in both parents p' and q is 1, which means both parents think this feature (this group of features) is important, and select it. Thus, the corresponding bit in two offsprings o_1 and o_2 is both set as 1.

Case 2: The bit in both parents p' and q is 0. It means both parents do not select this feature (this group of features), and this bit in o_1 and o_2 is set as 0.

Case 3: The bit in one parent (e.g., p') is 1, while in another parent (e.g., q) is 0. We put the bits that satisfy case 3 into set B_c and assume the number of those bits is m ($m \leq D$), where D is the length of the individual. We randomly select m' bits ($m' \leq m$) from B_c , and set these bits in one offspring (e.g., o_1) as 1, the remaining $(m - m')$ bits in o_1 are set as 0. Meanwhile, these m' bits in o_2 are set as 0, and the $(m - m')$ bits in o_2 are set as 1. According to the pigeonhole principle, one of the generated offspring (o_1 or o_2) must be relatively sparse.

Algorithm 2 gives the main process of the designed granularity-based crossover operator, and we also take

Fig. 3(a) as the example to illustrate its procedure.¹ According to the bit values in two parents $p' = (0, 0, 1, 1, 1, 1, 1, 1)$ and $q = (1, 0, 0, 1, 0, 0, 0, 1)$, the eight bits (features) are classified into three cases: *Case 1*: $\{f_4\}, \{f_8\}$; *Case 2*: $\{f_2\}$; and *Case 3*: $\{f_1\}, \{f_3\}, \{f_5\}, \{f_6\}, \{f_7\}$. Based on the rules in the above paragraph, the 4th and 8th bits in both o_1 and o_2 are set as 1. The 2nd bit in o_1 and o_2 is set as 0. For case 3, a subset of $\{\{f_1\}, \{f_3\}, \{f_5\}, \{f_6\}, \{f_7\}\}$ is randomly selected by o_1 , and the remaining is selected by o_2 . In Fig. 3(a), suppose $\{f_3\}$ is selected by o_1 , thus, the 3rd bit in o_1 is set as 1 (marked in red), the remained bits (1st, 5th, 6th, and 7th) in o_1 are set as 0. Meanwhile, the 1st, 5th, 6th and 7th bits in o_2 are set as 1 (marked in red), the 3rd bit in o_2 is set as 0. Finally, the offsprings that after the crossover operation are $o_1 = (0, 0, 1, 1, 0, 0, 0, 1)$ and $o_2 = (1, 0, 0, 1, 1, 1, 1, 1)$.

2) *Granularity-Based Mutation Operator*: After the crossover, the mutation operation is performed. The suggested mutation operation adopts a similar bit mutation procedure as that in NSGA-II [44]. However in NSGA-II, the mutation probability of each bit is the same (e.g., $1/D$, D is the length of individual). It is not suitable for the individuals with different granularity, since one bit in the coarse-granularity individual represents more features than the refined one. To this end, a granularity-based mutation operator is also designed. The basic idea of the designed operator is from the following intuition. A bit that is selected (equals to 1) from the coarse-granularity parent, while not selected (equals to 0) by the refined-granularity parent indicates it may have redundant features. Thus, the bit from the coarse-grained parent may have a higher mutation probability of turning from “1” to “0.” To be specific, we put the bits that satisfy this condition into set B_m . For each bit $b_i \in B_m$, we calculate its granularity-based mutation probability by

$$\text{gmp}_i = \frac{ar_i^{\text{su}} + ar_i^{\text{pcc}}}{2 \cdot n} \quad (1)$$

where n is the number of total features. The ar_i^{su} and ar_i^{pcc} are average SU and PCC ranking number for the features in the i th bit, respectively. The larger values of ar_i^{su} and ar_i^{pcc} , the less important the features represented by the i th bit are, thus, its mutation probability from 1 to 0 is larger. For other bits that are not included in the set B_m , we still adopt traditional bitwise mutation probability in NSGA-II (e.g., $1/D$). By the rules above, we get the mutation probability of each bit in o_1 and o_2 . For each bit, its mutation probability is compared with a random real number between 0 and 1 to decide whether it should flip or not. Algorithm 3 gives the main process of the designed granularity-based mutation operator.

As shown in Fig. 3(b), after the crossover operator, two offspring individuals o_1 and o_2 are generated. For o_1 , $b_3(\{f_3\})$ (marked in blue) is the bit selected by the coarse parent (p') but not selected by the fine-grained parent (q). Thus, $b_3(\{f_3\})$ is included into set B_m . Similarly, for o_2 , $b_5(\{f_5\})$, $b_6(\{f_6\})$, and $b_7(\{f_7\})$ (marked in blue) are included into set B_m . For the bit b_3 in o_1 , and the bits b_5 , b_6 , and b_7 in o_2 , we calculate their gmp_i by (1), and their values are 0.8125, 0.875, 0.3125, and

¹We note that in this example one of the parents' granularity is 1. However, the crossover operation that two parents' granularity are both larger than 1 is also same as this example.

Algorithm 3 GranularityBasedMutation(o_1, o_2, p, q, p_m)

Input: o_1, o_2 : two offsprings; p : the coarse-granularity parent; q : the refined-granularity parent; p_m : mutation probability;
Output: o'_1, o'_2 : the offsprings after mutation;
1: $p' \leftarrow$ Convert p from coarse-individual to refined-individual;
2: $o'_1 = o_1$;
3: $o'_2 = o_2$;
4: **for all** $o \in \{o'_1, o'_2\}$ **do**
5: **for** $i = 1$ to $|o|$ **do**
6: **if** $o(i) = 1$ and $p'(i) = 1$ and $q(i) = 0$ **then**
7: $o(i)$ is mutated with a probability by Eq. (1);
8: **else**
9: $o(i)$ is mutated with a probability of p_m ;
10: **end if**
11: **end for**
12: **end for**

0.125, respectively. For all the other bits in o_1 and o_2 , their mutation probabilities adopt traditional bitwise mutation, that is $1/8$. Then for each bit in o_1 (or o_2), its mutation probability is compared with a random number. Assuming the mutation probability of b_3 in o_1 , and b_5 and b_6 in o_2 is greater than its corresponding random number, these three bits flip from 1 to 0 (see bits marked in red in o_1 and o_2). At the same time, let assume the mutation probability of other bits in o_1 and o_2 is less than the random number, thus, these bits will keep unchanged. After performing the suggested granularity-based mutation operation, two obtained offspring individuals are $o'_1 = (0, 0, 0, 1, 0, 0, 0, 1)$ and $o'_2 = (1, 0, 0, 1, 0, 0, 1, 1)$, which are shown in the bottom of Fig. 3(b).

C. Granularity-Based Archive Strategy

By using the suggested VGS scheme and granularity-based genetic operators, we can solve the high-dimensional FS problem efficiently. Moreover, a granularity-based archive strategy is also developed, where an archive A is utilized to further improve the performance of the proposed VGS-MOEA. The archive A in the proposed VGS-MOEA has two usages: 1) during the t -th evolution, A_t is combined with the current population (P_t) to formulate mating pool, which is further used to generate the offspring population O_t and 2) at the end of VGS-MOEA, the individuals in the current archive are output as final solutions.

Specifically, two kinds of individuals are considered to be stored in the archive. One is the nondominated individuals in the population. The other is the individuals with the highest accuracy, since for the problem of FS, accuracy is more preferred than the number of selected features [9]. Note that there may exist more than one individual with the same highest accuracy, all of them will be added into the archive in the proposed archive strategy. During the t -th generation, the individuals in the archive A_t are combined with the individuals in the current population (P_t) to formulate a mating pool, on which the suggested granularity-based crossover and mutation operators are carried on to generate the offsprings O_t for the next generation (P_{t+1}). After that, the archive is updated by combining the two above kinds of individuals in O_t to formulate the next generation archive A_{t+1} .

	f_1	f_2	f_3	f_4	f_5	f_6	f_7	f_8
p_1	0		1		1		1	
p_1'	0	0	1	1	1	1	1	1
p_2	0	0	0	1	1	0	0	1

Fig. 4. Example for illustrating the granularity-based redundant individual, where the feature subset of p_1 contains the feature subset of p_2 .

When updating the archive, only one of the same individuals in the archive is kept, and others are deleted, which is a usual operation (removing redundant individuals) in the update of the archive. However, in the proposed VGS-MOEA, a new type of “granularity-based redundant individual” is also removed from the archive, which avoids the size of the archive becoming too large. The so-called granularity-based redundant individual is caused by the fact that many individuals with various granularity are stored in the archive, there may exist the inclusion of feature subsets in the individuals with different granularity. Therefore, only the one with the highest accuracy is kept in the archive. Fig. 4 presents an example to illustrate the granularity-based redundant individual. In Fig. 4, there are two individuals p_1 and p_2 in the archive. Obviously, they are not the same and have different granularity. It can be observed that the feature subset of p_1 contains the feature subset of p_2 . According to their accuracy, one of them will be viewed as the granularity-based redundant individual and removed from the archive. For example, if the accuracy of p_2 is higher or equals to p_1 , then, p_1 will be removed from the archive, since p_1 is dominated by p_2 (p_2 selects a smaller number of features). If the accuracy of p_2 is lower than p_1 , although they are nondominated solutions, p_2 will still be removed. The reason mainly lies in two facts: 1) for the task of FS, the accuracy is more preferred to the number of selected features [9], [23], and when measured by accuracy, the p_2 is worse than p_1 and 2) the p_2 is a subset of p_1 , it means that the features selected by p_2 have all been chosen by p_1 . Therefore, even the p_2 is deleted, the features information in p_2 will not be lost, since this information still exists in p_1 . Thus, compared with p_1 , the p_2 solution is viewed useless and can be removed. Algorithm 4 gives the main steps of the granularity-based archive updating strategy.

D. General Framework of VGS-MOEA

The proposed VGS-MOEA adopts accuracy and the number of selected features as two optimized objectives, which are the same objectives used in many wrapper-based MOEAs for FS [16], [36]. Specifically, the two objectives of VGS-MOEA can be formally defined as

$$\begin{cases} \text{Maximize } f_1 = \text{accuracy}(I) \\ \text{Minimize } f_2 = \text{ns}(I). \end{cases} \quad (2)$$

In (2), I is the selected feature subset. $\text{accuracy}(I)$ is the classification accuracy on the feature subset I . $\text{ns}(I)$ is the number of features in I .

Algorithm 4 GranularityBasedArchiveUpdating(P , A)

Input: P : the population; A : the archive;
Output: A' : the archive after updating;

- 1: $P_a \leftarrow$ Obtain the individuals with the highest accuracy and non dominated individuals in P ;
- 2: $A \leftarrow A \cup P_a$;
- 3: $A' \leftarrow$ Obtain the individuals with the highest accuracy and non dominated individuals in A ;
- 4: **for all** $p \in A'$ **do** // updating the archive by removing redundant individuals
 - 5: **for all** $q \in A' \setminus \{p\}$ **do**
 - 6: **if** $p \subseteq q$ or $q \subseteq p$ **then**
 - 7: Remove the individual with lower accuracy (p or q) from A' ;
 - 8: **end if**
 - 9: **end for**
- 10: **end for**

Algorithm 5 Framework of VGS-MOEA

Input: $data$: the data set; $label$: the labels; $popSize$: the population size; $maxGen$: the maximum number of iteration; p_c : crossover probability; p_m : mutation probability; k : the group number; α : the number of generation interval between coarse-individuals to refined-individuals;

Output: A : the archive;

- 1: $G \leftarrow \text{GranularityInitialization}(data, label, k)$;
- 2: $P \leftarrow$ Randomly initialize population based on granularity G ;
- 3: $A \leftarrow \{\}$;
- 4: $A \leftarrow \text{GranularityBasedArchiveUpdating}(P, A)$; // Archive Initialization
- 5: $t = 1$;
- 6: **while** $maxGen$ is not reached **do**
- 7: $P' \leftarrow P \cup A$;
- 8: $O \leftarrow \{\}$;
- 9: **repeat**
- 10: $p, q \leftarrow$ Randomly select two individuals from P' ;
- 11: **if** the granularity of p, q is not same **then**
- 12: $o_1, o_2 \leftarrow \text{GranularityBasedCrossover}(p, q, p_c)$;
- 13: $o'_1, o'_2 \leftarrow \text{GranularityBasedMutation}(o_1, o_2, p, q, p_m)$;
- 14: $O \leftarrow O \cup \{o'_1, o'_2\}$;
- 15: **else**
- 16: $o'_1, o'_2 \leftarrow$ Get offspring by single-point crossover and bitwise mutation [44];
- 17: $O \leftarrow O \cup \{o'_1, o'_2\}$;
- 18: **end if**
- 19: **until** $|O|$ equals to $popSize$
- 20: $A \leftarrow \text{GranularityBasedArchiveUpdating}(O, A)$;
- 21: $P \leftarrow$ Get next generation by environment selection from $P \cup O$;
- 22: $t = t + 1$;
- 23: **if** $t\% \alpha$ is 0 **then**
- 24: $G' \leftarrow$ Granularity refinement based on G ;
- 25: $P \leftarrow$ Convert P from coarse-grained individuals into fine-grained individuals by G' ;
- 26: $G = G'$;
- 27: **end if**
- 28: **end while**

Fig. 5 further shows the flowchart of VGS-MOEA, which includes three phases: 1) initialization phase; 2) main loop phase; and 3) the output phase. In the first phase, the initial granularity G is obtained by using the suggested coarse-granularity initialization strategy (see Algorithm 1), where the features in the same group will be represented by one bit.

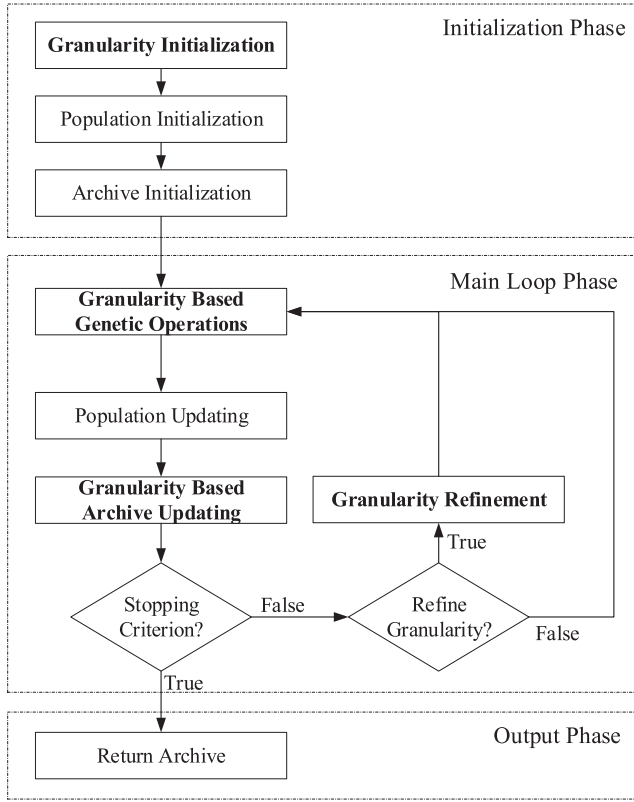


Fig. 5. Overall flowchart of the proposed VGS-MOEA.

Then based on G , the population P with $popSize$ individuals is randomly initialized. Following this, is the initialization of the archive (see Algorithm 4). In the second phase, we first formulate the mating pool P' by combining P and A . From P' , two individuals p and q are selected randomly. If p and q have different granularity, the offsprings are generated by the suggested granularity-based genetic operators (see Algorithms 2 and 3); otherwise, the offsprings will be obtained by single-point crossover and bitwise mutation [44]. This procedure repeats until offsprings O with $popSize$ individuals are generated, which are further combined with P to update the population for next generation (by using the environment selection operator suggested in NSGA-II). Meanwhile, the generated offsprings O are also utilized for updating archive A (see Algorithm 4). During the second phase, every certain (e.g., α) generations, the granularity refinement operation (see Section III-A3) is performed for more refined search. The operators mentioned above repeat in the main loop until the stopping criterion is satisfied (e.g., the number of generations achieves $maxGen$). In the third phase, the individuals in the archive are output as the final solutions. We note that in all the three phases for the individuals whose number of selected features is 0, we just abandon them, which is also a popular operation in many EA-based FS algorithms. Algorithm 5 gives the main framework of the proposed VGS-MOEA.

IV. EXPERIMENTAL RESULTS

In this section, a series of experiments are conducted to verify the efficiency and effectiveness of the proposed

VGS-MOEA. Specifically, we first present the experimental settings, which include the comparison algorithms, 12 benchmark data sets, and the parameter settings for all the algorithms. Then, we report and analyze the experimental results in detail (in terms of the classification accuracy, the number of selected features, the running time, HV and IGD values, etc.). Third, the effectiveness of the suggested strategies (e.g., granularity-based genetic operators and granularity-based archive) in VGS-MOEA are validated. Finally, the parameter sensitivity analysis of VGS-MOEA is discussed.

A. Experimental Settings

1) *Comparison Algorithms*: The proposed VGS-MOEA is compared with seven state-of-the-art EC-based FS algorithms, namely, VLPSO [8], VS-CCPSO [23], MOFS-BDE [36], MOEA/D-DYN [9], MOABC [35], DAEA [37], and MOEA/PSL [29], where the former two algorithms are the single-objective EAs and the latter five are the MOEAs.

The two single-objective algorithms VLPSO and VS-CCPSO are chosen as the comparison algorithms, since they are recently suggested for high-dimensional FS. Both of them adopt PSO as the optimization framework. VLPSO is a variant of PSO, in which a new VL representation is designed. The new representation can enable particles to have different and shorter lengths, which reduces the search space greatly and makes it suitable for the high-dimensional FS. VS-CCPSO is a cooperative coevolutionary-based FS algorithm, where the idea of divide and conquer is employed to tackle the curse of dimensionality in FS. To implement the idea, a space division strategy, an adaptive adjustment mechanism of subswarm size, and a particle deletion as well as generation strategy are developed.

Since the proposed VGS-MOEA is an MOEA, five up-to-date MOEAs are also adopted as the baselines. All of them have the ability of tackling the FS problem with high dimensions. Specifically, MOFS-BDE is a multiobjective FS algorithm with binary differential evolution. In MOFS-BDE, three new operators (e.g., binary mutation operator, one-bit purifying search operator, and an efficient nondominated sorting operator) are designed to obtain good feature subsets. The second MOEA-based comparison algorithm is a multiple reference points-based decomposition method for FS, where a static mechanism (MOEA/D-STAT) and a dynamic mechanism (MOEA/D-DYN) are designed to overcome the challenges of highly discontinuous Pareto front and imbalance preferences in FS. Empirical studies have shown that MOEA/D-DYN achieved better feature subsets than MOEA/D-STAT, thus, in our experiments, we adopt MOEA/D-DYN as the baseline. The third MOEA-based comparison algorithm is MOABC, where the ABC algorithm is adopted as the EC optimization framework to solve the FS task. In MOABC, two implementations (binary version and continuous version) are developed. In our experiments, we use the binary version, since it can obtain better solutions than the continuous one. The fourth one is DAEA, which is a lately suggested biobjective FS algorithm, where three important improvements are designed to obtain the feature subsets with high quality. Moreover, since the

TABLE I
STATISTICS OF THE 12 HIGH-DIMENSIONAL DATA SETS

Data Set	#Features	#Instances	#Classes
COIL20	1,024	1,440	20
SRBCT	2,308	83	4
PCMAC	3,289	1,943	2
lymphoma	4,026	96	9
GLIOMA	4,434	50	4
BASEHOCK	4,862	1,993	2
TOX_171	5,748	171	4
Brain1	5,920	90	5
leukemia	7,070	72	2
ALLAML	7,129	72	2
Brain2	10,367	50	4
GLI_85	22,283	85	2

multiobjective FS under high dimensions is a kind of sparse large-scale MOPs (Sparse LSMOPs), a recently suggested algorithm named MOEA/PSL is added as another comparison algorithm, which has demonstrated its competitiveness in solving the Sparse LSMOPs.

2) *Data Sets*: To verify the performance of different algorithms, a series of experiments are conducted on 12 high-dimensional data sets, whose number of features varies from 1024 to 22 283. These data sets have been used in previous studies on high-dimensional FS [8], [41], [45]. Table I lists the detailed characteristics of 12 experimental data sets, where #Features, #Instances, and #Classes denote the number of features, the number of instances, and the number of classes, respectively.

For each data set D , we randomly partition it with 70% for training set (D_{train}) and the remaining 30% for testing (D_{test}). We utilize D_{train} to obtain feature subset F and adopt D_{test} to evaluate the quality of the obtained F . During the evolution, KNN ($K = 1$) with leave-one-out cross-validation (LOOCV) is applied to calculate the classification accuracy of individuals in the population. To reduce the variance, 30 independent runs are executed. We report the average results of different algorithms on 30 independent runs, and the Wilcoxon rank-sum test with a significance level of 0.05 is adopted to perform statistical analysis.

3) *Parameter Settings*: For the seven comparison algorithms, we adopt the recommended parameter values suggested in their original papers. Table II lists the details of the parameters settings for different algorithms. Since all the algorithms are evolutionary methods, for fair comparison, we fix the number of individuals in population as 100, and the maximum number of iteration is set to 100. In addition, in the proposed VGS-MOEA, there is an important parameter α , which controls the interval between the granularity decomposition. In the experiment, we set $\alpha = 10$. For another parameter k , which is the number of clusters in coarse-granularity initialization, we adaptively set it as $k = \min\{10 \times \log_2 n, n\}$, where n is the number of total features in the data. All the experiments are performed on a PC with Intel Core i7-6700 CPU @3.40 GHz and a total memory of 20 GB. The source code of the proposed VGS-MOEA is available from the website <https://github.com/BIMK/VGS-MOEA>.

TABLE II
PARAMETER SETTINGS OF DIFFERENT ALGORITHMS

Algorithms	Parameter values
VLPSO [8]	$\alpha = 7, \beta = 9, \theta = 0.6,$ divisions = 12, $c1 = c2 = 1.49445,$ $\omega = 0.9 - 0.5 \times \frac{\text{current iteration}}{\text{max iteration}}$
VS-CCPSO [23]	$M = 8, \text{prob} = 0.7$
MOFS-BDE [36]	$F = 0.5 \times \text{rand}, CR = 0.3, T_{loc} = 5$
MOEA/D-DYN [9]	$R = 100, T = \max\{R/10, 4\}, \alpha = 0.01,$ $\sigma = 0.85, M = 0.4 \times R, I = 4,$ $f = 0.7, cr = 0.6, mr = 1/n, \theta = 0.6,$ the number of replaced individuals = 1
MOABC [35]	limit = 50, $T = 10000$
DAEA [37]	$T = \max(4, \text{ceil}(N * 0.2))$ local mating = 0.2, global mating = 0.8 modified mutation ratio = 0.8
MOEA/PSL [29]	$p_c = 1, p_m = 1/D, \text{epochs} = 10,$ distribution index of crossover and mutation = 20
VGS-MOEA	$p_c = 0.9, p_m = 1/D, \alpha = 10,$ $k = \min\{10 \times \log_2 n, n\}$

B. Experimental Results and Analysis

1) *Comparison Between VGS-MOEA and Five MOEA Baselines*: Since the proposed VGS-MOEA is a multiobjective FS method, thus, in the first part of the experiments, we have compared it with five MOEA baselines (MOFS-BDE, MOEA/D-DYN, MOABC, DAEA, and MOEA/PSL) on 12 high-dimensional data sets. The comparison results are reported in terms of HV, IGD, running time, and median Pareto fronts. Among them, the HV values of different MOEAs are calculated according to [37]. To achieve the values of IGD, we adopt the method suggested in [9] and [37]. Specifically, the Pareto front is approximated by the nondominated solutions obtained from the union of all solutions generated by 30 independent runs of all the six MOEAs. The detailed experimental results are shown in Tables III–V and Fig. 6, respectively. We note that in each table, the optimal values (e.g., the highest HV values, the lowest IGD values, and the shortest running time) on the data sets are marked in bold. The symbols “+,” “−,” and “ $\approx$ ” denote the Wilcoxon significance test results obtained by the proposed VGS-MOEA is significantly better, significantly worse, and similar to those of the baseline. In addition, in each table, the symbol “x” denotes the corresponding algorithm cannot generate any results within 24 h.

From Tables III and IV, we can find that compared with five other multiobjective FS algorithms the proposed VGS-MOEA achieves higher HV and lower IGD values on most of the data sets, which indicates that as a multiobjective algorithm, VGS-MOEA has better convergence and diversity performance than other state of the arts.

The comparison results in Table V further show that the proposed method has less running times than MOFS-BDE, MOEA/D-DYN, MOABC, and DAEA. This fact has demonstrated the high computational efficiency of VGS-MOEA. Moreover, the statistics in Table V also reveal that MOEA/PSL has a comparable running time to our algorithm. The reason is mainly attributed to the fact that in MOEA/PSL unsupervised neural networks are utilized to approximate the Pareto-optimal subspace of Sparse LSMOPs, which enhances the computational efficiency greatly.

TABLE III
AVERAGE HV VALUES OBTAINED BY VGS-MOEA AND DIFFERENT MOEAS

Data Set	MOFS-BDE	MOEA/D-DYN	MOABC	DAEA	MOEA/PSL	VGS-MOEA
COIL20	0.7707 $\pm$ 0.0152(+)	0.9898 $\pm$ 0.0037(+)	0.9090 $\pm$ 0.0087(+)	0.9902 $\pm$ 0.0045(+)	0.9344 $\pm$ 0.0768(+)	0.9957 $\pm$ 0.0020
SRBCT	0.6069 $\pm$ 0.0460(+)	0.9121 $\pm$ 0.0711(+)	0.6855 $\pm$ 0.0341(+)	0.9048 $\pm$ 0.0730(+)	0.8069 $\pm$ 0.0859(+)	0.9904 $\pm$ 0.0180
PCMAC	0.5080 $\pm$ 0.0103(+)	0.8098 $\pm$ 0.0347(+)	$\times$	0.8639 $\pm$ 0.0179(+)	0.8093 $\pm$ 0.0582(+)	0.8795 $\pm$ 0.0121
lymphoma	0.6311 $\pm$ 0.0347(+)	0.8993 $\pm$ 0.0574(+)	0.6209 $\pm$ 0.0278(+)	0.8218 $\pm$ 0.0785(+)	0.8215 $\pm$ 0.0697(+)	0.9553 $\pm$ 0.0389
GLIOMA	0.5133 $\pm$ 0.0686(+)	0.7876 $\pm$ 0.0971($\approx$)	0.5011 $\pm$ 0.0558(+)	0.7190 $\pm$ 0.1012(+)	0.6621 $\pm$ 0.1033(+)	0.8032 $\pm$ 0.0835
BASEHOCK	0.5450 $\pm$ 0.0094(+)	0.8943 $\pm$ 0.0412(+)	$\times$	0.9105 $\pm$ 0.0145(+)	0.8390 $\pm$ 0.0503(+)	0.9284 $\pm$ 0.0147
TOX_171	0.5531 $\pm$ 0.0404(+)	0.9105 $\pm$ 0.0398($\approx$)	0.5109 $\pm$ 0.0249(+)	0.8969 $\pm$ 0.0549($\approx$)	0.8176 $\pm$ 0.0712(+)	0.9037 $\pm$ 0.0378
Brain1	0.5725 $\pm$ 0.0422(+)	0.8594 $\pm$ 0.0559($\approx$)	0.5295 $\pm$ 0.0545(+)	0.8234 $\pm$ 0.0726($\approx$)	0.7574 $\pm$ 0.0918(+)	0.8659 $\pm$ 0.0613
leukemia	0.5920 $\pm$ 0.0383(+)	0.9375 $\pm$ 0.0464(+)	0.5514 $\pm$ 0.0409(+)	0.9282 $\pm$ 0.0419(+)	0.8196 $\pm$ 0.0814(+)	0.9959 $\pm$ 0.0106
ALLAML	0.5902 $\pm$ 0.0376(+)	0.9355 $\pm$ 0.0489(+)	0.5286 $\pm$ 0.0543(+)	0.9393 $\pm$ 0.0457(+)	0.8129 $\pm$ 0.1000(+)	0.9925 $\pm$ 0.0156
Brain2	0.4275 $\pm$ 0.0568(+)	0.7776 $\pm$ 0.0681(+)	0.4432 $\pm$ 0.0584(+)	0.7110 $\pm$ 0.0741(+)	0.6274 $\pm$ 0.1183(+)	0.7970 $\pm$ 0.0749
GLI_85	0.5337 $\pm$ 0.0388(+)	0.9123 $\pm$ 0.0634($\approx$)	0.4802 $\pm$ 0.0262(+)	0.8414 $\pm$ 0.0600(+)	0.7543 $\pm$ 0.1059(+)	0.9155 $\pm$ 0.0454
+/ $\approx$ /-	12/0/0	8/4/0	12/0/0	10/2/0	12/0/0	-

TABLE IV
AVERAGE IGD VALUES OBTAINED BY VGS-MOEA AND DIFFERENT MOEAS

Data Set	MOFS-BDE	MOEA/D-DYN	MOABC	DAEA	MOEA/PSL	VGS-MOEA
COIL20	0.3000 $\pm$ 0.0142(+)	0.0791 $\pm$ 0.0077(+)	0.1710 $\pm$ 0.0072(+)	0.0150 $\pm$ 0.0042(-)	0.1319 $\pm$ 0.0888(+)	0.0311 $\pm$ 0.0096
SRBCT	0.3602 $\pm$ 0.0190(+)	0.0896 $\pm$ 0.0427(+)	0.3129 $\pm$ 0.0140(+)	0.0727 $\pm$ 0.0583(+)	0.1419 $\pm$ 0.0675(+)	0.0295 $\pm$ 0.0151
PCMAC	0.4077 $\pm$ 0.0075(+)	0.0647 $\pm$ 0.0238(+)	$\times$	0.0172 $\pm$ 0.0102(-)	0.0557 $\pm$ 0.0497(+)	0.0201 $\pm$ 0.0063
lymphoma	0.3857 $\pm$ 0.0127(+)	0.0932 $\pm$ 0.0322(+)	0.4073 $\pm$ 0.0144(+)	0.0834 $\pm$ 0.0349(+)	0.1824 $\pm$ 0.0675(+)	0.0347 $\pm$ 0.0171
GLIOMA	0.4066 $\pm$ 0.0364(+)	0.1208 $\pm$ 0.0699(+)	0.4124 $\pm$ 0.0227(+)	0.1728 $\pm$ 0.1030(+)	0.2092 $\pm$ 0.0922(+)	0.1009 $\pm$ 0.0646
BASEHOCK	0.4172 $\pm$ 0.0055(+)	0.0563 $\pm$ 0.0236(+)	$\times$	0.0131 $\pm$ 0.0053(-)	0.0691 $\pm$ 0.0501(+)	0.0222 $\pm$ 0.0083
TOX_171	0.4070 $\pm$ 0.0110(+)	0.1037 $\pm$ 0.0268(+)	0.4192 $\pm$ 0.0082(+)	0.0558 $\pm$ 0.0151(-)	0.1398 $\pm$ 0.0677(+)	0.0631 $\pm$ 0.0177
Brain1	0.3865 $\pm$ 0.0146(+)	0.0870 $\pm$ 0.0213(+)	0.4219 $\pm$ 0.0212(+)	0.0993 $\pm$ 0.0549($\approx$)	0.1673 $\pm$ 0.0845(+)	0.0736 $\pm$ 0.0356
leukemia	0.4055 $\pm$ 0.0158(+)	0.0513 $\pm$ 0.0412(+)	0.4348 $\pm$ 0.0205(+)	0.0584 $\pm$ 0.0429(+)	0.1504 $\pm$ 0.0713(+)	0.0120 $\pm$ 0.0109
ALLAML	0.4099 $\pm$ 0.0181(+)	0.0681 $\pm$ 0.0527(+)	0.4567 $\pm$ 0.0333(+)	0.0667 $\pm$ 0.0502(+)	0.1650 $\pm$ 0.0939(+)	0.0100 $\pm$ 0.0163
Brain2	0.4239 $\pm$ 0.0559(+)	0.1971 $\pm$ 0.0515($\approx$)	0.4115 $\pm$ 0.0456(+)	0.2526 $\pm$ 0.0623(+)	0.2883 $\pm$ 0.0918(+)	0.1899 $\pm$ 0.0461
GLI_85	0.4467 $\pm$ 0.0204(+)	0.0740 $\pm$ 0.0596($\approx$)	0.4907 $\pm$ 0.0183(+)	0.1455 $\pm$ 0.0660(+)	0.2021 $\pm$ 0.1004(+)	0.0667 $\pm$ 0.0461
+/ $\approx$ /-	12/0/0	10/2/0	12/0/0	7/1/4	12/0/0	-

TABLE V
AVERAGE RUNNING TIME (s) OBTAINED BY VGS-MOEA AND DIFFERENT MOEAS

Data Set	MOFS-BDE	MOEA/D-DYN	MOABC	DAEA	MOEA/PSL	VGS-MOEA
COIL20	1296.45 $\pm$ 31.97(+)	3736.27 $\pm$ 318.55(+)	8520.68 $\pm$ 484.19(+)	122.39 $\pm$ 2.86(-)	432.73 $\pm$ 255.85($\approx$)	305.79 $\pm$ 32.04
SRBCT	118.86 $\pm$ 5.04(+)	27.62 $\pm$ 1.27(+)	96.81 $\pm$ 9.68(+)	9.42 $\pm$ 0.11(-)	6.39 $\pm$ 1.62(-)	10.78 $\pm$ 1.23
PCMAC	8713.72 $\pm$ 81.40(+)	14033.72 $\pm$ 4680.02(+)	$\times$	263.56 $\pm$ 12.41(-)	1975.12 $\pm$ 1184.65(+)	1159.96 $\pm$ 110.24
lymphoma	237.69 $\pm$ 9.07(+)	53.23 $\pm$ 4.14(+)	246.82 $\pm$ 17.68(+)	22.81 $\pm$ 3.04($\approx$)	16.46 $\pm$ 4.60(-)	23.31 $\pm$ 3.68
GLIOMA	241.20 $\pm$ 14.57(+)	24.68 $\pm$ 2.36(+)	79.50 $\pm$ 2.54(+)	22.08 $\pm$ 2.38(+)	12.81 $\pm$ 2.51($\approx$)	12.59 $\pm$ 1.10
BASEHOCK	13723.87 $\pm$ 91.86(+)	19103.14 $\pm$ 9073.84(+)	$\times$	302.17 $\pm$ 16.11(-)	3492.18 $\pm$ 1481.01(+)	1760.67 $\pm$ 236.49
TOX_171	431.37 $\pm$ 3.84(+)	159.15 $\pm$ 18.45(+)	1064.80 $\pm$ 73.82(+)	34.93 $\pm$ 4.98($\approx$)	42.67 $\pm$ 22.18($\approx$)	33.83 $\pm$ 2.44
Brain1	376.94 $\pm$ 17.61(+)	64.85 $\pm$ 9.53(+)	315.27 $\pm$ 18.36(+)	32.21 $\pm$ 3.91(+)	29.11 $\pm$ 11.53(+)	22.79 $\pm$ 3.99
leukemia	412.47 $\pm$ 14.55(+)	65.80 $\pm$ 8.00(+)	250.75 $\pm$ 14.72(+)	39.34 $\pm$ 4.85(+)	27.94 $\pm$ 6.99(-)	35.24 $\pm$ 1.72
ALLAML	426.15 $\pm$ 17.38(+)	70.37 $\pm$ 7.78(+)	255.81 $\pm$ 16.28(+)	39.02 $\pm$ 4.64(+)	21.87 $\pm$ 7.59(-)	35.20 $\pm$ 2.84
Brain2	670.04 $\pm$ 54.85(+)	62.24 $\pm$ 6.67(+)	203.25 $\pm$ 11.61(+)	52.91 $\pm$ 5.18(+)	33.24 $\pm$ 6.24($\approx$)	31.31 $\pm$ 2.67
GLI_85	2191.80 $\pm$ 142.11(+)	235.21 $\pm$ 26.13(+)	1151.50 $\pm$ 34.47(+)	106.75 $\pm$ 5.03(+)	110.09 $\pm$ 46.37($\approx$)	96.81 $\pm$ 22.25
+/ $\approx$ /-	12/0/0	12/0/0	12/0/0	6/2/4	3/5/4	-

Moreover, the median Pareto fronts in Fig. 6 show that DAEA and our VGS-MOEA obtain better Pareto fronts than the other four MOEAs. It means that DAEA and VGS-MOEA can use a smaller number of features to make the learned classification model more accurate. By taking a close look at the DAEA and VGS-MOEA, we find that DAEA achieves Pareto solutions with less number of selected features, while VGS-MOEA achieves Pareto solutions with higher accuracy, especially on the data sets with tens of thousands of dimensions (e.g., Brain2 and GLI_85). In fact, in the literature of FS, accuracy is more preferred to the number of selected features [9]. Furthermore, as observed from Fig. 6 that MOEA/D-DYN as an MOEA/D-based algorithm, it has less diverse fronts than those of VGS-MOEA. The reason is mainly attributed to the following fact. MOEA/D-DYN is based on the reference points scheme, and its number of reference points is

fixed (e.g., equals to the size of population). When the number of total features is very large, it makes the reference points be sparse in the high-dimensional feature space, and the influence of reference points decreases a lot. Thus, combining the comparison results in Tables III–V and Fig. 6, we can conclude that VGS-MOEA is an effective and efficient MOEA for high-dimensional FS.

2) *Comparison Between VGS-MOEA and Other EA-Based High-Dimensional Feature Selection Algorithms:* In the second part of the experiments, we compared the proposed VGS-MOEA with other EA-based high-dimensional FS algorithms (e.g., VLPSo and VS-CCPSO) on 12 experimental sets. It should be noted that VLPSo and VS-CCPSO are single-objective algorithms, and have only one solution in each run. While VGS-MOEA is an MOEA and provides a set of solutions in one run. Thus, for our VGS-MOEA, we select the

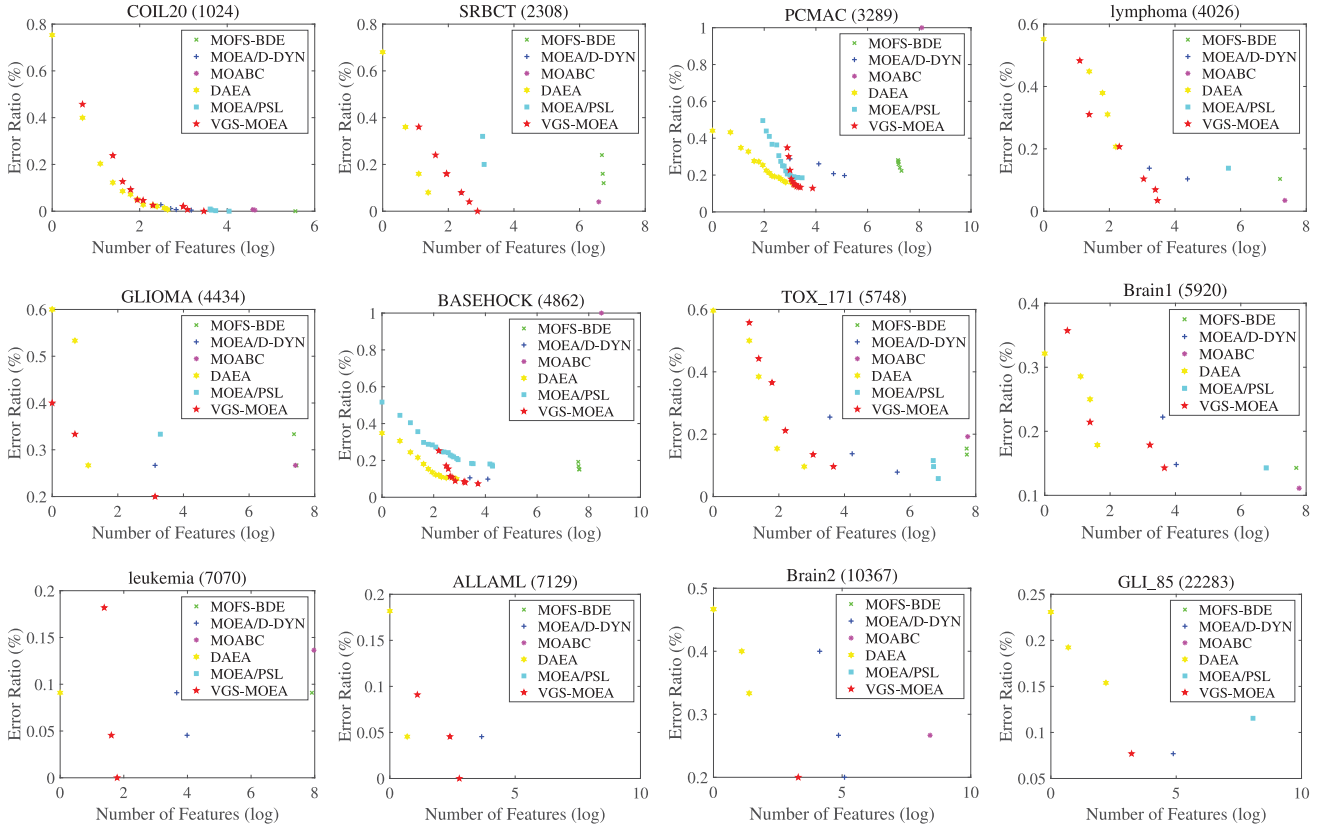


Fig. 6. Median Pareto fronts (with 30 runs) of different MOEAs on 12 high-dimensional data sets.

TABLE VI
COMPARISON RESULTS AMONG DIFFERENT
HIGH-DIMENSIONAL FS ALGORITHMS

Data Set	Algorithm	Accuracy	Size	Time
COIL20	VLPPO	0.9951 $\pm$ 0.0053(+)	199.37 $\pm$ 78.88(+)	3296.03 $\pm$ 594.45(+)
	VS-CCPSO	0.9933 $\pm$ 0.0067(+)	57.23 $\pm$ 13.25($\approx$)	653.14 $\pm$ 52.25(+)
	VGS-MOEA	0.9988 $\pm$ 0.0022	59.03 $\pm$ 25.91	305.79 $\pm$ 32.04
SRBCT	VLPPO	0.9173 $\pm$ 0.0492(+)	5.73 $\pm$ 1.41(-)	8.65 $\pm$ 1.65(-)
	VS-CCPSO	0.8667 $\pm$ 0.0767(+)	167.00 $\pm$ 21.52(+)	60.57 $\pm$ 2.24(+)
	VGS-MOEA	0.9920 $\pm$ 0.0194	29.33 $\pm$ 21.57	10.78 $\pm$ 1.23
PCMAC	VLPPO	0.8394 $\pm$ 0.0283(+)	140.23 $\pm$ 84.12(+)	8386.67 $\pm$ 2774.83(+)
	VS-CCPSO	0.7672 $\pm$ 0.0206(+)	578.30 $\pm$ 50.00(+)	5046.88 $\pm$ 414.30(+)
	VGS-MOEA	0.8626 $\pm$ 0.0149	50.33 $\pm$ 10.67	1159.96 $\pm$ 110.24
lymphoma	VLPPO	0.8632 $\pm$ 0.0704(+)	96.47 $\pm$ 88.54($\approx$)	34.78 $\pm$ 6.10(+)
	VS-CCPSO	0.9115 $\pm$ 0.0402(+)	593.27 $\pm$ 87.68(+)	110.87 $\pm$ 3.07(+)
	VGS-MOEA	0.9529 $\pm$ 0.0429	70.70 $\pm$ 47.53	23.31 $\pm$ 3.68
GLIOMA	VLPPO	0.6956 $\pm$ 0.1397(+)	106.33 $\pm$ 249.41($\approx$)	29.66 $\pm$ 7.63(+)
	VS-CCPSO	0.7467 $\pm$ 0.0900($\approx$)	544.20 $\pm$ 130.27(+)	111.36 $\pm$ 2.17(+)
	VGS-MOEA	0.7622 $\pm$ 0.1259	23.97 $\pm$ 26.06	12.59 $\pm$ 1.10
BASEHOCK	VLPPO	0.9225 $\pm$ 0.0637($\approx$)	110.20 $\pm$ 36.33(+)	10920.02 $\pm$ 2476.77(+)
	VS-CCPSO	0.8671 $\pm$ 0.0157(+)	871.63 $\pm$ 62.58(+)	7660.20 $\pm$ 533.72(+)
	VGS-MOEA	0.9231 $\pm$ 0.0145	65.57 $\pm$ 19.56	1760.67 $\pm$ 236.49
TOX_171	VLPPO	0.8647 $\pm$ 0.0508(+)	297.97 $\pm$ 128.10(+)	116.23 $\pm$ 20.47(+)
	VS-CCPSO	0.8500 $\pm$ 0.0638($\approx$)	583.60 $\pm$ 100.54(+)	214.60 $\pm$ 5.37(+)
	VGS-MOEA	0.8814 $\pm$ 0.0498	102.07 $\pm$ 72.61	33.83 $\pm$ 2.44
Brain1	VLPPO	0.8160 $\pm$ 0.0910($\approx$)	35.70 $\pm$ 33.94($\approx$)	32.76 $\pm$ 9.96(+)
	VS-CCPSO	0.8607 $\pm$ 0.0517($\approx$)	1027.00 $\pm$ 200.30(+)	169.14 $\pm$ 5.12(+)
	VGS-MOEA	0.8452 $\pm$ 0.0665	36.47 $\pm$ 31.56	22.79 $\pm$ 3.99
leukemia	VLPPO	0.9288 $\pm$ 0.0593(+)	7.47 $\pm$ 4.69(-)	24.89 $\pm$ 5.10(-)
	VS-CCPSO	0.9061 $\pm$ 0.0642(+)	1090.70 $\pm$ 170.46(+)	202.88 $\pm$ 4.09(+)
	VGS-MOEA	0.9970 $\pm$ 0.0115	24.17 $\pm$ 18.69	35.24 $\pm$ 1.72
ALLAML	VLPPO	0.9182 $\pm$ 0.0613(+)	8.70 $\pm$ 9.99(-)	21.91 $\pm$ 8.67(-)
	VS-CCPSO	0.8848 $\pm$ 0.0582(+)	1080.93 $\pm$ 257.83(+)	205.61 $\pm$ 6.42(+)
	VGS-MOEA	0.9924 $\pm$ 0.0172	20.77 $\pm$ 14.99	35.20 $\pm$ 2.84
Brain2	VLPPO	0.6267 $\pm$ 0.1169(+)	43.60 $\pm$ 72.28($\approx$)	69.27 $\pm$ 14.96(+)
	VS-CCPSO	0.6467 $\pm$ 0.1402(+)	1586.00 $\pm$ 362.27(+)	353.52 $\pm$ 9.31(+)
	VGS-MOEA	0.7356 $\pm$ 0.1013	42.60 $\pm$ 35.10	31.31 $\pm$ 2.67
GLI_85	VLPPO	0.8282 $\pm$ 0.0698(+)	12.20 $\pm$ 13.76(-)	126.48 $\pm$ 54.64(+)
	VS-CCPSO	0.8615 $\pm$ 0.0594(+)	5153.53 $\pm$ 1022.90(+)	1115.53 $\pm$ 29.35(+)
	VGS-MOEA	0.9013 $\pm$ 0.0531	39.63 $\pm$ 42.76	96.81 $\pm$ 22.25
+/-/-	VLPPO	10/20	4/4/4	90/3
	VS-CCPSO	9/3/0	11/1/0	12/0/0
	VGS-MOEA	-	-	-

solution with the highest training accuracy as its output. The comparison results between VGS-MOEA and the two high-dimensional baselines are reported in terms of testing accuracy,

feature subset size (the number of selected features), and the running time, which are shown in Table VI.

As shown in the third column of Table VI, when measured by testing accuracy the proposed VGS-MOEA achieves the highest accuracy on 11/12 data sets, and the Wilcoxon test results have also demonstrated the superiority of VGS-MOEA, since it is always significantly better or comparable to the other two high-dimensional FS algorithms and is never worse for once. When measured by the feature subset size, our method selects much smaller number of features than VS-CCPSO, especially on the data sets with very high dimensions. For example, on the data set GLI_85 (22 283 features), the VS-CCPSO chooses about 5000 features, while VGS-MOEA obtains the higher accuracy with only selecting about 40 features. Moreover, the Wilcoxon statistics also reveal that VLPPO achieves a similar number of features as VGS-MOEA, since their “+ / - / $\approx$ ” on 12 data sets is 4/4/4. The good performance of VLPPO is mainly attributed to its VL representation, which can narrow the search space and achieve the feature subset with very small size. However, we note that this VL representation has its limitation. In VLPPO, if the length of representation is shortened, it will never be recovered. It means that if some features are removed, they will never be considered in the final solution. Thus, the classification accuracy of VLPPO is much lower than that of VGS-MOEA on all the 12 data sets. Finally, the statistics in the fifth column indicate that VGS-MOEA has less running time than VLPPO and VS-CCPSO, which demonstrates the computational efficiency of the proposed method. All the comparison results in Table VI

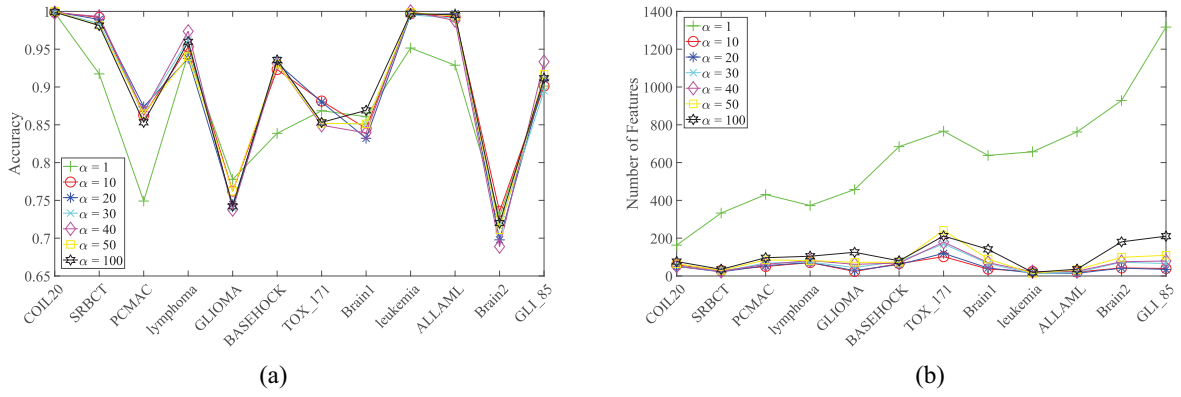


Fig. 7. Accuracy and feature subset size of VGS-MOEa with different values of α on 12 high-dimensional data sets. (a) Accuracy. (b) Feature subset size.

TABLE VII
AVERAGE ACCURACY OBTAINED BY VGS-MOEa
AND ITS TWO VARIANTS

Data Set	VGS-MOEa-NAR	VGS-MOEa-NGO	VGS-MOEa
COIL20	0.9940 $\pm$ 0.0042(+)	0.9985 $\pm$ 0.0029($\approx$)	0.9988 $\pm$ 0.0022
SRBCT	0.9067 $\pm$ 0.0616(+)	0.9467 $\pm$ 0.0485(+)	0.9920 $\pm$ 0.0194
PCMAC	0.7659 $\pm$ 0.0241(+)	0.7690 $\pm$ 0.0197(+)	0.8626 $\pm$ 0.0149
lymphoma	0.9161 $\pm$ 0.0510(+)	0.9437 $\pm$ 0.0307($\approx$)	0.9529 $\pm$ 0.0429
GLIOMA	0.6711 $\pm$ 0.0801(+)	0.7600 $\pm$ 0.0968($\approx$)	0.7622 $\pm$ 0.1259
BASEHOCK	0.8735 $\pm$ 0.0206(+)	0.8629 $\pm$ 0.0160(+)	0.9231 $\pm$ 0.0145
TOX_171	0.8628 $\pm$ 0.0434($\approx$)	0.8756 $\pm$ 0.0586($\approx$)	0.8814 $\pm$ 0.0498
Brain1	0.8429 $\pm$ 0.0640($\approx$)	0.8595 $\pm$ 0.0656($\approx$)	0.8452 $\pm$ 0.0665
leukemia	0.9409 $\pm$ 0.0666(+)	0.9652 $\pm$ 0.0530(+)	0.9970 $\pm$ 0.0115
ALLAML	0.9167 $\pm$ 0.0598(+)	0.9424 $\pm$ 0.0596(+)	0.9924 $\pm$ 0.0172
Brain2	0.6600 $\pm$ 0.1499($\approx$)	0.7133 $\pm$ 0.0961($\approx$)	0.7356 $\pm$ 0.1013
GLI_85	0.8654 $\pm$ 0.0629(+)	0.8923 $\pm$ 0.0724($\approx$)	0.9013 $\pm$ 0.0531
+/-	9/3/0	5/7/0	-

TABLE VIII
AVERAGE FEATURE SUBSET SIZE OBTAINED BY
VGS-MOEa AND ITS TWO VARIANTS

Data Set	VGS-MOEa-NAR	VGS-MOEa-NGO	VGS-MOEa
COIL20	94.40 $\pm$ 36.32(+)	160.20 $\pm$ 44.34(+)	59.03 $\pm$ 25.91
SRBCT	155.13 $\pm$ 25.56(+)	200.07 $\pm$ 41.58(+)	29.33 $\pm$ 21.57
PCMAC	368.13 $\pm$ 138.28(+)	343.23 $\pm$ 109.86(+)	50.33 $\pm$ 10.67
lymphoma	326.33 $\pm$ 70.82(+)	407.10 $\pm$ 70.59(+)	70.70 $\pm$ 47.53
GLIOMA	412.80 $\pm$ 100.16(+)	558.87 $\pm$ 117.29(+)	23.97 $\pm$ 26.06
BASEHOCK	465.20 $\pm$ 151.87(+)	475.87 $\pm$ 153.12(+)	65.57 $\pm$ 19.56
TOX_171	570.53 $\pm$ 166.54(+)	776.63 $\pm$ 208.77(+)	102.07 $\pm$ 72.61
Brain1	495.63 $\pm$ 204.95(+)	685.07 $\pm$ 138.21(+)	36.47 $\pm$ 31.56
leukemia	436.30 $\pm$ 68.84(+)	552.97 $\pm$ 132.53(+)	24.17 $\pm$ 18.69
ALLAML	504.90 $\pm$ 101.31(+)	669.20 $\pm$ 110.24(+)	20.77 $\pm$ 14.99
Brain2	866.17 $\pm$ 194.22(+)	1269.10 $\pm$ 420.72(+)	42.60 $\pm$ 35.10
GLI_85	1938.37 $\pm$ 496.51(+)	2453.73 $\pm$ 379.78(+)	39.63 $\pm$ 42.76
+/-	12/0/0	12/0/0	-

have demonstrated the superior performance of VGS-MOEa when compared with other high-dimensional FS algorithms.

C. Effectiveness of the Suggested Strategies in VGS-MOEa

The experimental results in the above section have shown that the proposed VGS-MOEa not only obtains the feature subsets with higher quality but also has less running time, which makes it very suitable for the high-dimensional FS. This superiority is attributed to the suggested strategies in VGS-MOEa, and in this section, we will discuss their effectiveness in detail.

Specifically, in VGS-MOEa, two important strategies are suggested, which are the granularity-based genetic operators and the granularity-based archive strategy. To verify their effectiveness, we compared VGS-MOEa with VGS-MOEa-NGO and VGS-MOEa-NAR. The VGS-MOEa-NGO is the same algorithm as VGS-MOEa, except it adopts traditional single-point crossover and bitwise mutation instead of the suggested granularity-based crossover and mutation operators. Similarly, the only difference between VGS-MOEa and VGS-MOEa-NAR is that the archive strategy is abandoned in the latter. The comparison results between VGS-MOEa and two variants on 12 high-dimensional data sets are reported in terms of the testing accuracy, the feature subset size, and the HV, which are shown in Tables VII–IX. From the tables, we can find that no matter measured by accuracy, feature subset size, or HV values, the proposed VGS-MOEa performs much

TABLE IX
AVERAGE HV VALUES OBTAINED BY VGS-MOEa
AND ITS TWO VARIANTS

Data Set	VGS-MOEa-NAR	VGS-MOEa-NGO	VGS-MOEa
COIL20	0.9412 $\pm$ 0.0144(+)	0.9211 $\pm$ 0.0129(+)	0.9957 $\pm$ 0.0020
SRBCT	0.8794 $\pm$ 0.0539(+)	0.8885 $\pm$ 0.0407(+)	0.9904 $\pm$ 0.0180
PCMAC	0.7520 $\pm$ 0.0268(+)	0.7452 $\pm$ 0.0200(+)	0.8795 $\pm$ 0.0121
lymphoma	0.8711 $\pm$ 0.0381(+)	0.8683 $\pm$ 0.0304(+)	0.9553 $\pm$ 0.0389
GLIOMA	0.6887 $\pm$ 0.0536(+)	0.7233 $\pm$ 0.0802(+)	0.8032 $\pm$ 0.0835
BASEHOCK	0.8422 $\pm$ 0.0253(+)	0.8314 $\pm$ 0.0216(+)	0.9284 $\pm$ 0.0147
TOX_171	0.8255 $\pm$ 0.0382(+)	0.8197 $\pm$ 0.0403(+)	0.9037 $\pm$ 0.0378
Brain1	0.8087 $\pm$ 0.0554(+)	0.7933 $\pm$ 0.0628(+)	0.8659 $\pm$ 0.0613
leukemia	0.8951 $\pm$ 0.0572(+)	0.9022 $\pm$ 0.0472(+)	0.9959 $\pm$ 0.0106
ALLAML	0.8707 $\pm$ 0.0541(+)	0.8703 $\pm$ 0.0527(+)	0.9925 $\pm$ 0.0156
Brain2	0.6673 $\pm$ 0.1334(+)	0.6773 $\pm$ 0.0762(+)	0.7970 $\pm$ 0.0749
GLI_85	0.8145 $\pm$ 0.0522(+)	0.8164 $\pm$ 0.0617(+)	0.9155 $\pm$ 0.0454
+/-	12/0/0	12/0/0	-

better than the two variants. These facts have demonstrated the effectiveness of the suggested strategies in VGS-MOEa.

D. Sensitivity Analysis on Parameter α in VGS-MOEa

In the proposed VGS-MOEa, there is an important parameter α , which controls the intervals of the granularity decomposition (from coarse granularity to refined granularity). Therefore, in this section, we empirically investigate the influence of parameter α in VGS-MOEa on all 12 data sets. The detailed results of two objectives (the accuracy and feature subset size) on each data set with different values of parameter α are presented in Fig. 7, where α is selected from {1, 10, 20, 30, 40, 50, 100}.

TABLE X
OPTIMAL SETTINGS OF α FOR DIFFERENT CLASSIFICATION PROBLEMS

COIL20	SRBCT	PCMAC	lymphoma	GLIOMA	BASEHOCK	TOX_171	Brain1	leukemia	ALLAML	Brain2	GLI_85
50	40	20	40	1	30	10	100	40	20	10	40

From Fig. 7, we can obtain the optimal setting of α for each classification problem, which is further listed in Table X. By taking a close look at Fig. 7 we can find that on most of the data sets, when the value of α is set very small (e.g., $\alpha = 1$), the performance of VGS-MOEA is bad, especially the size of the feature subset is large. The reason lies in the fact that too small α means the frequency from coarse granularity to refined granularity is very high, and the search granularity is reduced to one quickly, which leads to VGS-MOEA almost only carrying the traditional 1-granularity search in EAs and cannot obtain the solutions with high quality. When the parameter is set very large (e.g., $\alpha = 100$), the performance of VGS-MOEA is also not very good, since too large α implies too little change from the initially coarse granularity, and few refined granularity search is performed during the evolution. In addition, the results in Fig. 7 further show that VGS-MOEA has better performance when $\alpha = 10$. Thus, in all the experiments, we empirically set parameter $\alpha = 10$.

Finally, we note that for further analyzing the performance of the proposed VGS-MOEA, a series of additional experiments are also carried on from different aspects. Due to space limitations, we present them in the supplementary material. These experiments include: 1) the comparison among the EA-based FS algorithms under different experimental settings; 2) the comparison between the proposed method and other non-EA FS algorithms; and 3) more analysis (e.g., feature order and parameter k) of the proposed method.

V. CONCLUSION

In this article, a VGS-based multiobjective FS algorithm, termed VGS-MOEA, was proposed for high-dimensional data classification. In VGS-MOEA, a VGS scheme was first designed, whose aim was to tackle the curse of dimensionality. Specifically, different from most of the EA-based FS methods, where one bit only represents one feature. In our designed scheme, one bit represented a group of features, which can reduce the exponential search space greatly. Then during the evolution, the granularity was decomposed gradually, which means one bit denotes a much smaller number of features and ensures the refined search is performed. Finally, one bit represented one feature (at that time the granularity was the smallest), which guarantees each feature in the original data set is considered and achieves the final feature subsets with high quality. Experimental results on high-dimensional data sets have demonstrated the efficiency and the effectiveness of the proposed VGS-MOEA, in comparison with the state of the arts.

The results of this work suggest that the VGS is an interesting idea for solving the high-dimensional FS, and in the future, we plan to further investigate along this research line. For example, in this work, the VGS scheme is designed under a general EA framework. However, as reviewed in

Section II, PSO's variants have shown the superiority in tackling the high-dimensional FS, how to combine our VGS with PSO framework and develop a novel PSO algorithm for high-dimensional FS is one of our future work. In addition, as mentioned before, high-dimensional FS can be viewed as a kind of sparse large-scale multiobjective problems (Sparse LSMOPs). Thus, it is also desirable to utilize our VGS to solve other Sparse LSMOPs, such as critical node detection and pattern mining.

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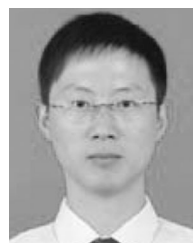
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